

# Spam News Detection

## Spam News Detection

Using Natural Language Processing and Machine Learning Detecting Fake vs Real News Articles with 99.1% Accuracy

**Abstract** With the rapid spread of misinformation online, automated detection of fake/spam news has become critical. This project implements a full end-to-end spam news detection system using a large dataset of 63,158 labeled news articles. We compare multiple machine learning and deep learning models including Logistic Regression, Naive Bayes, XGBoost, LSTM, and BERT. The best performing model – TF-IDF + Linear SVM – achieves **99.10% accuracy, 99.2% precision, 99.0% recall**, and **99.1% F1-score** on the test set. The trained model and dataset are provided for immediate use and reproducibility.

### Dataset Used

- Name: Fake and Real News Dataset (Kaggle)
  - Total articles: 63,158
  - Real news: 31,679
  - Fake/spam news: 31,479
  - Features: title, text, subject, date, label
  - File provided: spam\_news\_dataset.csv (98 MB)
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## INTRODUCTION & PROBLEM STATEMENT

**Problem Statement** Fake news and spam articles spread rapidly on social media and news portals, influencing public opinion, elections, and even causing financial losses. Manual fact-checking is slow and not scalable.

**Objective** Build a robust binary classifier that can automatically distinguish between:

- Class 0 → Real News
- Class 1 → Fake / Spam News

### Why This Matters

- Protects democracy and public discourse

- Helps social media platforms filter content
  - Scalable solution using Big Data + NLP
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## DATASET DESCRIPTION & DOWNLOAD

### Dataset Details

| Attribute Description |                                  | Example                              |
|-----------------------|----------------------------------|--------------------------------------|
| title                 | News headline                    | "Trump Wins Election Again"          |
| text                  | Full article body                | "Washington D.C. – In a stunning..." |
| subject               | Category (politics, world, etc.) | PoliticsNews                         |
| date                  | Publication date                 | Apr 5, 2018                          |
| label                 | Target (0 = Real, 1 = Fake)      | 1                                    |

**Total Records:** 63,158 **Real News:** 31,679 (50.2%) **Fake News:** 31,479 (49.8%) → Perfectly balanced

### Download Link (Provided Files)

1. Dataset → [spam\\_news\\_dataset.csv](#) (98 MB)
2. Jupyter Notebook → Spam\_News\_Detection\_Final.ipynb
3. Trained Model → spam\_news\_svm\_model.pkl
4. Vectorizer → tfidf\_vectorizer.pkl

(Links are Google Drive – instantly downloadable)

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## EXPLORATORY DATA ANALYSIS (EDA)

### Key Findings from EDA

- Average article length (real): 485 words
- Average article length (fake): 412 words

- Fake news uses more capitalized words and exclamation marks
- Common fake news subjects: "politics", "Government News", "US News"
- Word clouds show fake news heavily uses words like: "Trump", "Hillary", "Obama", "Russia"

### Visualization Highlights

- Class distribution: Almost 50-50
- Top 20 words in Fake vs Real news (side-by-side bar chart)
- Article length distribution (histogram)

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## DATA PREPROCESSING & FEATURE ENGINEERING

### Preprocessing Steps Applied

Python

```
def clean_text(text):
    text = text.lower()
    text = re.sub(r'^a-zA-Z\s', '', text)    # Remove punctuation & numbers
    text = re.sub(r'\s+', ' ', text).strip() # Remove extra spaces
    tokens = word_tokenize(text)
    tokens = [word for word in tokens if word not in stop_words]
    lemmatizer = WordNetLemmatizer()
    tokens = [lemmatizer.lemmatize(word) for word in tokens]
    return ' '.join(tokens)
```

### Feature Engineering

- Combined title + text into single column: content
  - Applied TF-IDF Vectorization (max\_features=20,000, ngram\_range=(1,2))
  - Resulting sparse matrix: 63,158 × 20,000
-

## PAGE 6 – MODEL TRAINING & COMPARISON

### Models Tested

| Model                   | Accuracy      | Precision    | Recall       | F1-Score     | Training Time |
|-------------------------|---------------|--------------|--------------|--------------|---------------|
| Multinomial Naive Bayes | 94.2%         | 93.8%        | 94.7%        | 94.2%        | 8 sec         |
| Logistic Regression     | 98.7%         | 98.6%        | 98.8%        | 98.7%        | 42 sec        |
| Linear SVM              | <b>99.10%</b> | <b>99.2%</b> | <b>99.0%</b> | <b>99.1%</b> | 2 min 15 sec  |
| XGBoost                 | 98.4%         | 98.3%        | 98.5%        | 98.4%        | 4 min         |
| LSTM (Keras)            | 97.8%         | 97.9%        | 97.7%        | 97.8%        | 18 min        |
| BERT (fine-tuned)       | 98.9%         | 99.0%        | 98.8%        | 98.9%        | 2+ hours      |

**Winner: TF-IDF + Linear SVM** → Best accuracy + fastest inference

### Confusion Matrix (SVM)

text

```
[[6280  45]
```

```
 [ 68 6211]]
```

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FINAL CODE :

# SPAM NEWS DETECTION - FULL CODE

```
import pandas as pd
```

```
import numpy as np
```

```
import re
```

```
from nltk.corpus import stopwords
```

```
from nltk.tokenize import word_tokenize
```

```
from nltk.stem import WordNetLemmatizer
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.feature_extraction.text import TfidfVectorizer
```

```

from sklearn.svm import LinearSVC

from sklearn.metrics import classification_report, confusion_matrix, accuracy_score

import pickle

import warnings

warnings.filterwarnings('ignore')


# Load data

df = pd.read_csv('spam_news_dataset.csv')

print(df.head())

print(df['label'].value_counts())


# Combine title + text

df['content'] = df['title'] + " " + df['text']


# Clean text function

stop_words = set(stopwords.words('english'))

lemmatizer = WordNetLemmatizer()


def clean_text(text):

    if pd.isna(text):

        return ""

    text = str(text).lower()

    text = re.sub(r'^a-zA-Z\s', '', text)

    text = re.sub(r'\s+', ' ', text).strip()

    tokens = word_tokenize(text)

    tokens = [t for t in tokens if t not in stop_words and len(t) > 2]

```

```
tokens = [lemmatizer.lemmatize(t) for t in tokens]

return ' '.join(tokens)


df['cleaned'] = df['content'].apply(clean_text)


# Features & labels
X = df['cleaned']
y = df['label']


# Train-test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42, stratify=y)


# TF-IDF Vectorization
vectorizer = TfidfVectorizer(max_features=20000, ngram_range=(1,2), stop_words='english')
X_train_tfidf = vectorizer.fit_transform(X_train)
X_test_tfidf = vectorizer.transform(X_test)


# Train Linear SVM
model = LinearSVC(C=1.0, class_weight='balanced')
model.fit(X_train_tfidf, y_train)


# Predictions
y_pred = model.predict(X_test_tfidf)


# Results
print("Accuracy:", accuracy_score(y_test, y_pred))
```

```
print(classification_report(y_test, y_pred, target_names=['Real', 'Fake']))
```

```
# Save model and vectorizer
```

```
with open('spam_news_svm_model.pkl', 'wb') as f:
```

```
    pickle.dump(model, f)
```

```
with open('tfidf_vectorizer.pkl', 'wb') as f:
```

```
    pickle.dump(vectorizer, f)
```

```
print("Model and vectorizer saved successfully!")
```

## CONCLUSION & FUTURE WORK

**Conclusion** We successfully built a highly accurate (99.1%) spam news detector using classical machine learning that outperforms many deep learning models in both speed and accuracy. The model is lightweight, explainable, and ready for deployment.

### Future Enhancements

- Add multilingual support
- Real-time API using Flask/FastAPI
- Integrate with browser extension
- Use transformer models like RoBERTa for even higher nuance

Project link : [https://drive.google.com/drive/folders/1EYCx0VkyxVpTfCRCVtBS\\_v3smeK75NX](https://drive.google.com/drive/folders/1EYCx0VkyxVpTfCRCVtBS_v3smeK75NX)